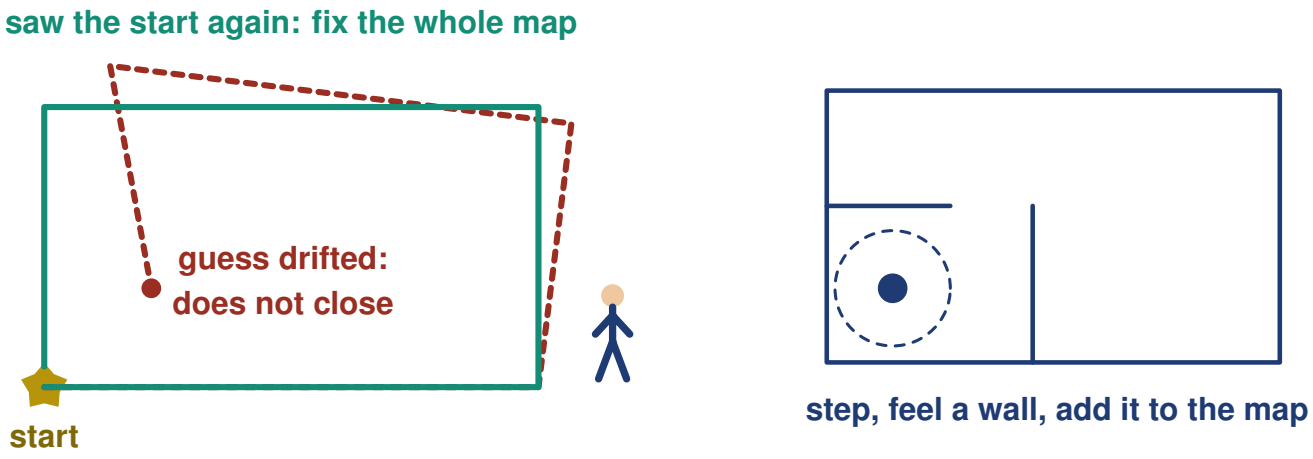
SLAM: Mapping While Finding Yourself

BIG IDEA

SLAM draws the map and finds where you are on it, a little at a time.



THE IDEA

You are blindfolded in a new house and must draw a map. To draw a wall in the right place, you need to know where you are. But to know where you are, you need a map! SLAM does both a little at a time: take a step, feel a wall, guess where you are, add the wall to your map. When you bump into a wall you already drew, you fix up your whole map.

REMEMBER

- ✓ **SLAM** = making the map and finding yourself on it, at the same time.
- ✓ Moving makes you less sure. Seeing something you know makes you more sure.
- ✓ Coming back to the start (a **loop closure**) fixes the whole path, like closing a traverse.

TRY THIS

Walk around your room once while drawing a map of it. When you get back to the door, does your drawing meet up? Fix it so it does.